Inventory management system for diamonds, colored stones, and jewelries

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What problem we are trying to solve ? (1)

Companies in the diamond and precious stone sector face several operational challenges:

- Paper-based processes that create delays and errors.
- Fragmented spreadsheets create duplicates, slow searches, and inventory errors.
- Limited traceability (certificates, provenance) increases audit and compliance risk.

What problem we are trying to solve ? (2)

Lot Concept

- either a single diamond or gemstone
- or a single piece of jewelry.

Each lot is characterized by:

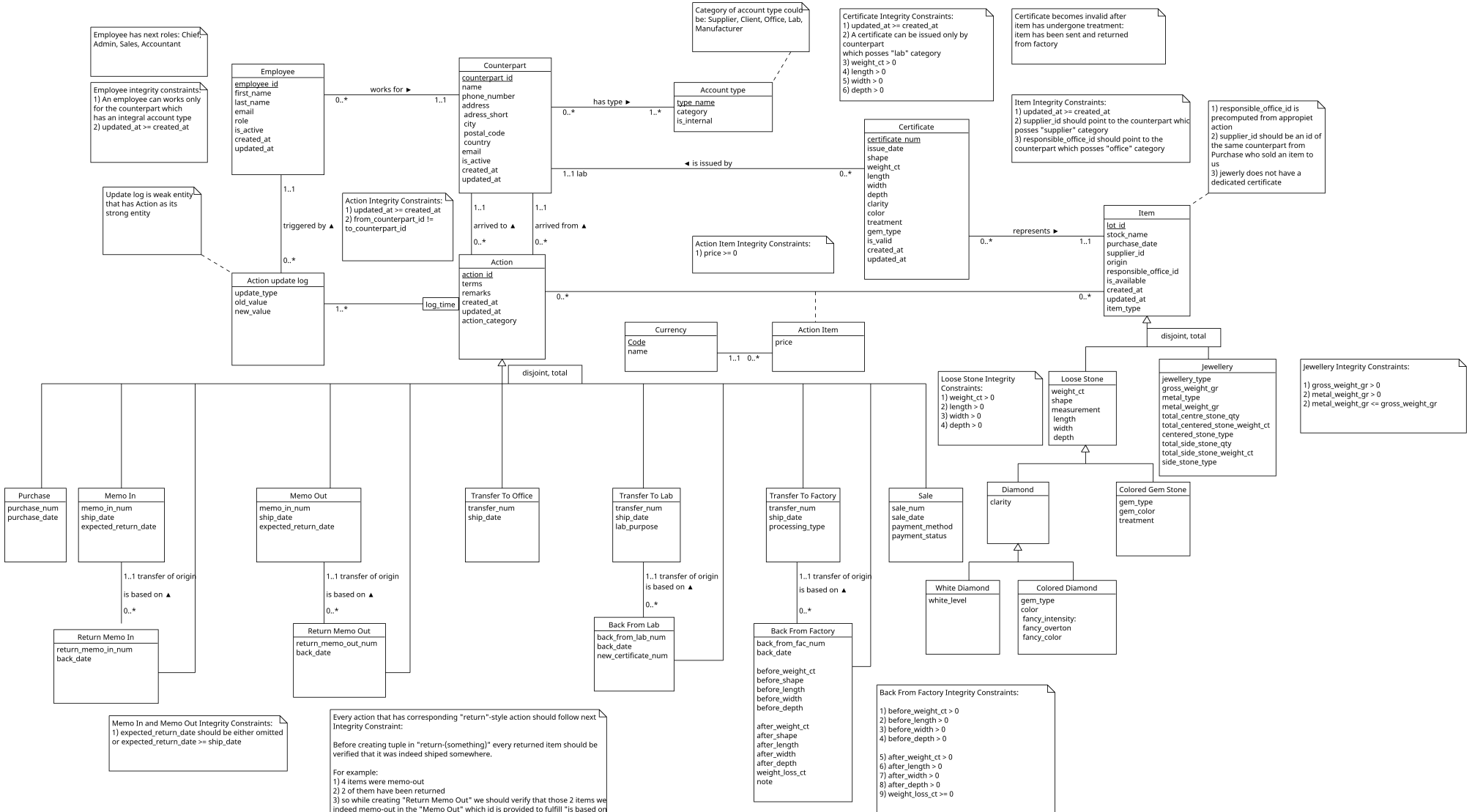
- A controlled status, belonging to a predefined and validated set of states
- A current location, which may be an internal office or an external partner
- An item category (diamond, gemstone, or jewelry)
- A purchase date and, when applicable, a sale date
- A linked counterparty (supplier, client, laboratory, or manufacturer)
- Financial information associated with the lot through commercial documents

What problem we are trying to solve ? (3)

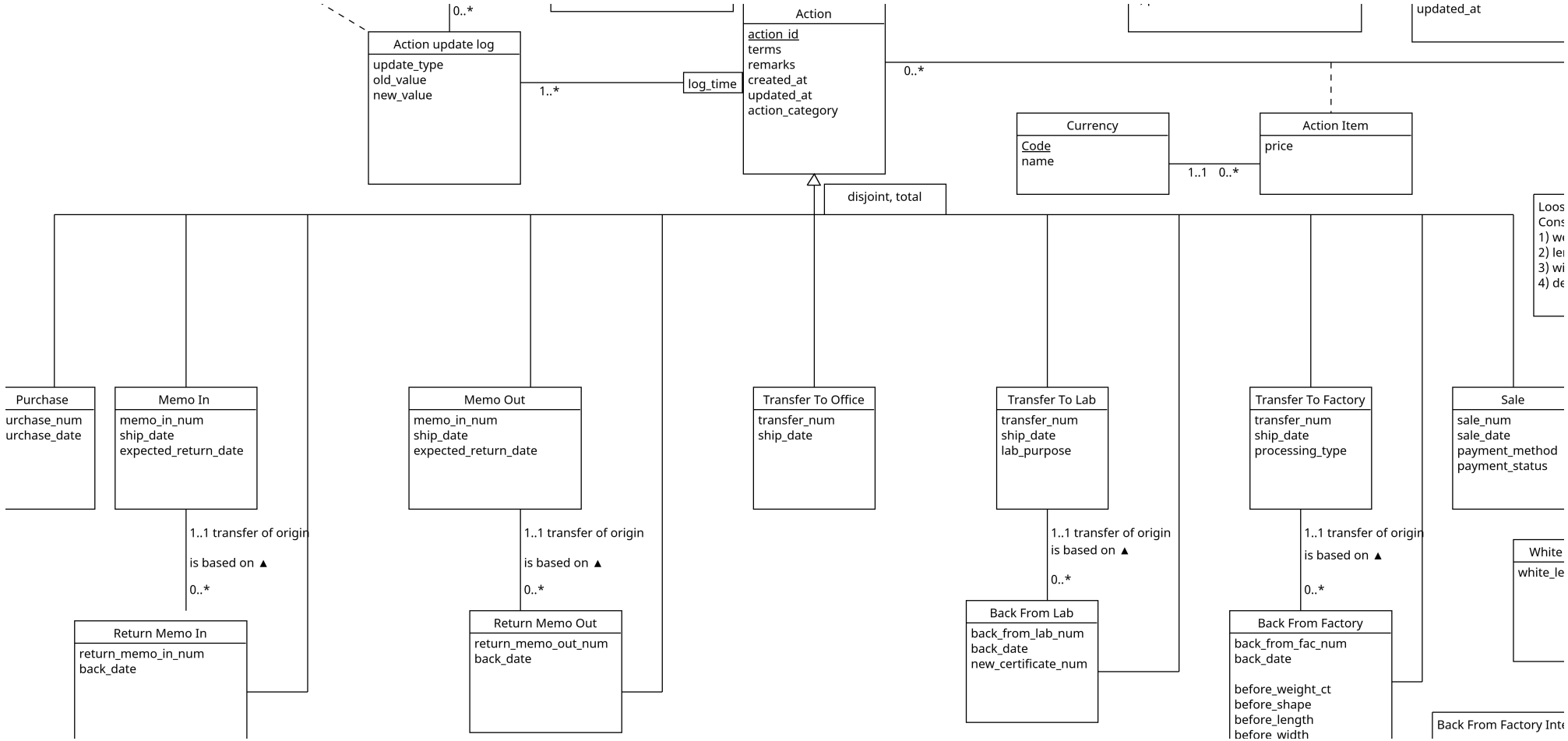
Inventory Lifecycle is driven by real world operations

- Purchase Note
- Memo In
- Return Memo In
- Memo Out
- Return Memo Out
- Transfer records
- Return Transfer records
- Invoice

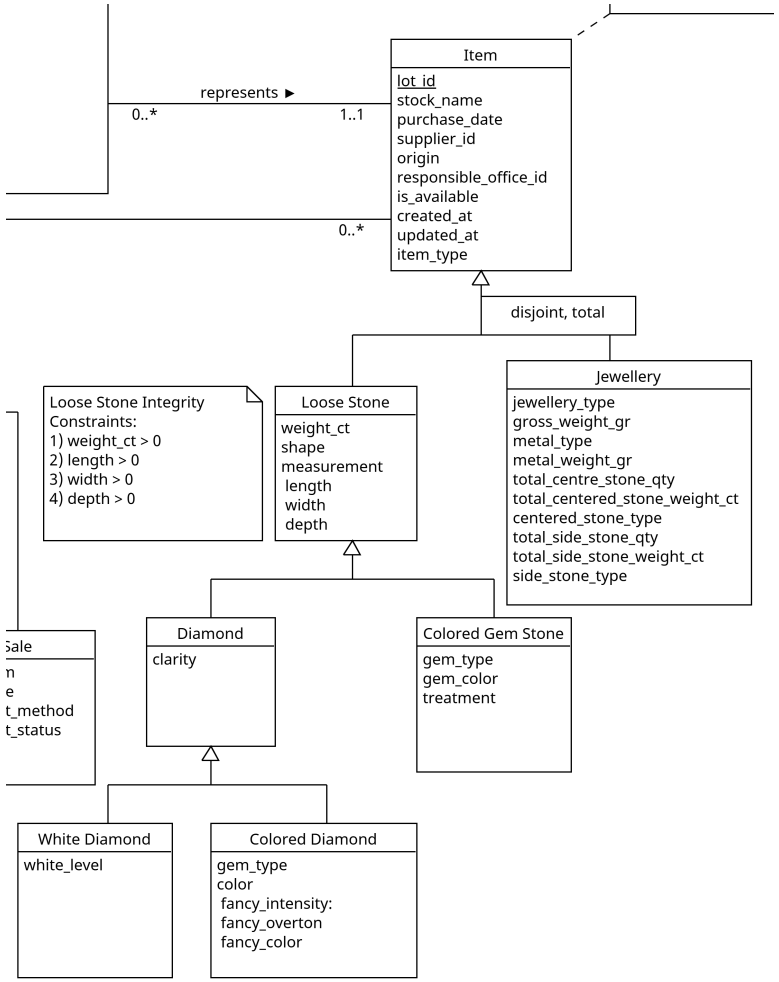
Conceptual schema (1)



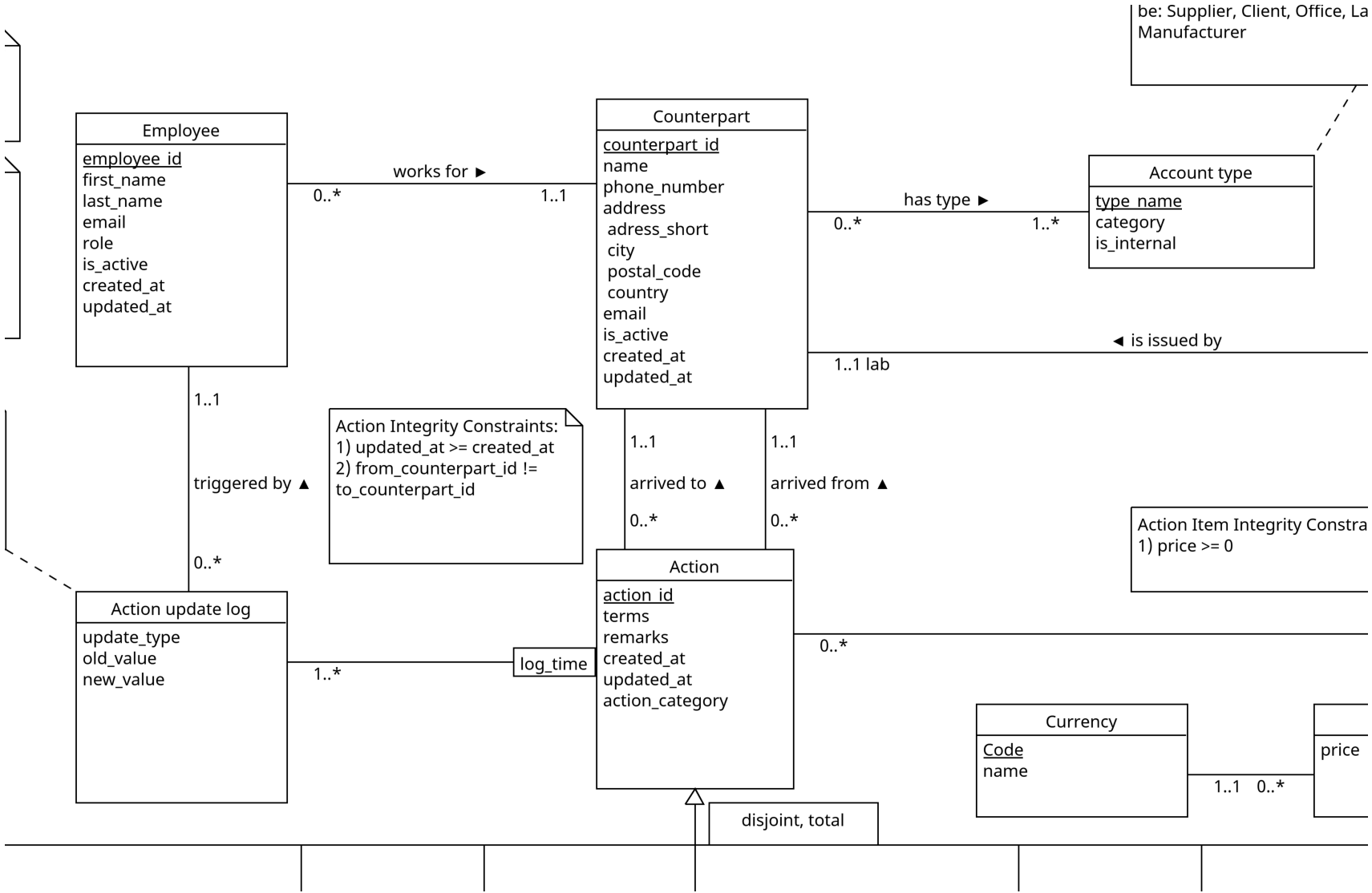
Conceptual schema (2)



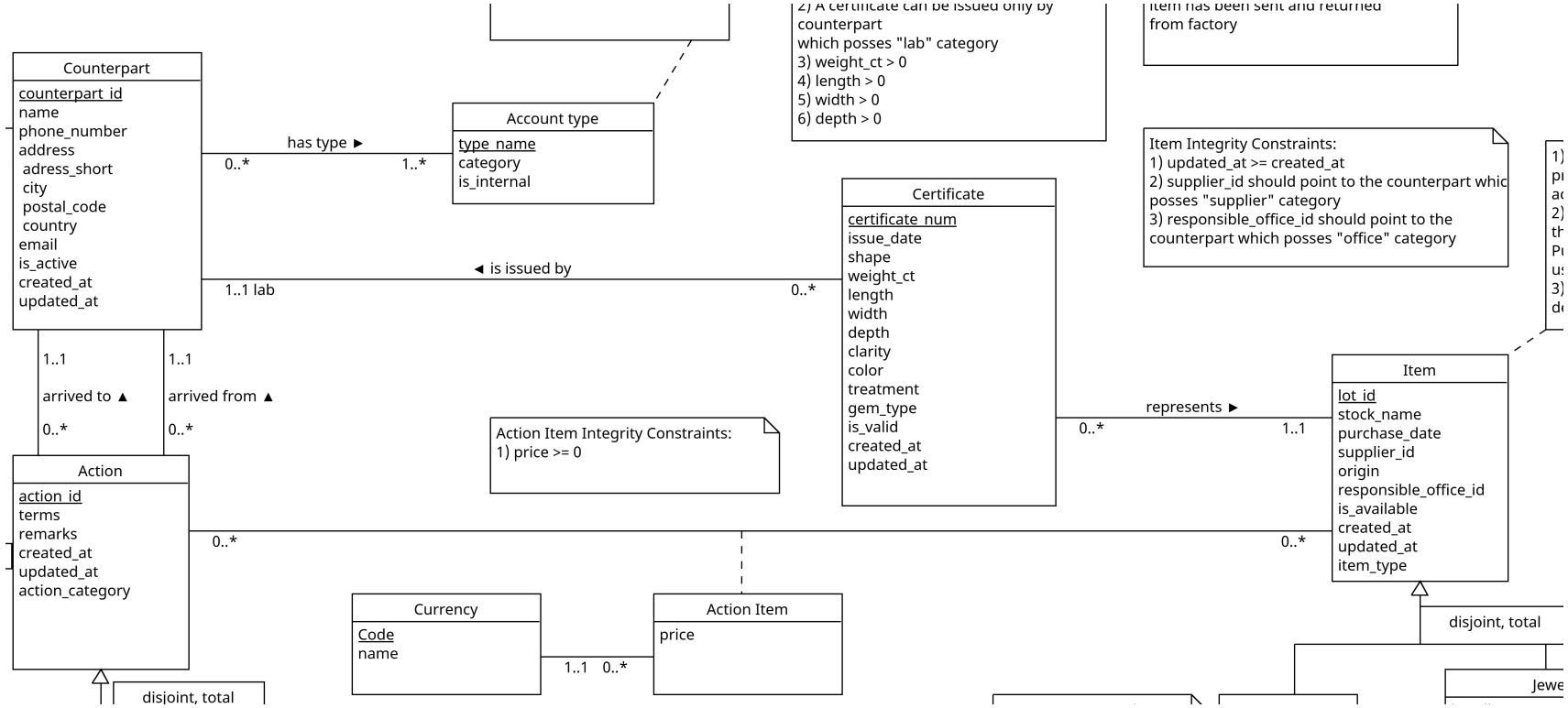
Conceptual schema (3)



Conceptual schema (4)



Conceptual schema (5)



Relation schema (1)

`item` related relations

item (**lot_id**, stock_name, purchase_date, supplier_id, origin, responsible_office_id, created_at, updated_at, is_available, item_type)

`supplier_id` references `counterpart.counterpart_id` NOT NULL

`responsible_office_id` references `counterpart.counterpart_id` NOT NULL

loose_stone (**lot_id**, weight_ct, shape, length, width, depth)

`lot_id` references `item.lot_id`

white_diamond (**lot_id**, white_level, clarity)

`lot_id` references `loose_stone.lot_id`

Relation schema (2)

`action` related relations

action (**action_id**, from_counterpart_id, to_counterpart_id, terms, remarks, created_at, updated_at, action_category)

`from_counterpart_id` references `counterpart.counterpart_id` NOT NULL

`to_counterpart_id` references `counterpart.counterpart_id` NOT NULL

purchase (**action_id**, purchase_num, purchase_date)

`action_id` references `action.action_id`

transfer_to_office (**action_id**, transfer_num, ship_date)

`action_id` references `action.action_id`

sale (**action_id**, sale_num, sale_date, payment_method, payment_status)

`action_id` references `action.action_id`

Relation schema (3)

return-style `action`

transfer_to_lab (**action_id**, transfer_num, ship_date, lab_purpose)
`action_id` references `action.action_id`
back_from_lab (**action_id**, orig_transfer_id, back_from_lab_num, back_date, new_certificate_num)
`action_id` references `action.action_id`
`orig_transfer_id` references `transfer_to_lab.action_id` NOT NULL
`new_certificate_num` references `certificate.certificate_num` NOT NULL

Relation schema (4)

Link between `action` and `item`

action_item (**action_id, lot_id**, price, currency_code)

`action_id` references `action.action_id`

`lot_id` references `item.lot_id`

`currency_code` references `currency.code` NOT NULL

Relation schema (5)

Link between `employee` and `action`

employee (**employee_id**, counterpart_id, first_name, last_name, email, role, is_active, created_at, updated_at)
`counterpart_id` references `counterpart.counterpart_id` NOT NULL
action_update_log (**log_time, action_id**, employee_id, update_type, old_value, new_value, log_time)
`action_id` references `action.action_id`
`employee_id` references `employee.employee_id` NOT NULL

Relation schema (6)

`counterpart` related relations

counterpart (**counterpart_id**, name, phone_number, address_short, city, postal_code, country, email, is_active, created_at, updated_at)

account_type (**type_name**, category, is_internal)

counterpart_account_type (**counterpart_id, type_name**)

`counterpart_id` references `counterpart.counterpart_id`

`type_name` references `account_type.type_name`

Relation schema (7)

`certificate`

certificate(**certificate_num**, lot_id, lab_id, issue_date, shape, weight_ct, length, width, depth, clarity, color, treatment, gem_type, is_valid, created_at, updated_at)

`lot_id` references `item.lot_id` NOT NULL

`lab_id` references `counterpart.counterpart_id` NOT NULL

SQL (1)

'Inventory by type' view (part 1)

```sql
CREATE OR REPLACE VIEW inventory_by_type AS
SELECT CASE
           WHEN wd.lot_id IS NOT NULL THEN 'White Diamond'
           WHEN cd.lot_id IS NOT NULL THEN 'Colored Diamond'
           WHEN cgs.lot_id IS NOT NULL THEN 'Colored Gemstone'
           WHEN j.lot_id IS NOT NULL THEN 'Jewelry'
           ELSE 'Unknown'
           END                                                AS item_tp,
       COUNT(*)                                               AS total_count,
       SUM(CASE WHEN i.is_available = TRUE THEN 1 ELSE 0 END)  AS available_count,
       SUM(CASE WHEN i.is_available = FALSE THEN 1 ELSE 0 END) AS unavailable_count,
       ROUND(SUM(COALESCE(ls.weight_ct, 0))::NUMERIC, 2)       AS total_weight_ct,
       ROUND(AVG(COALESCE(ls.weight_ct, 0))::NUMERIC, 2)       AS avg_weight_ct
FROM -- part 2 next slide
```

SQL (2)

'Inventory by type' view (`FROM` part)

```sql
CREATE OR REPLACE VIEW inventory_by_type AS
SELECT
  -- part 1 prev slide
  FROM item i
       LEFT JOIN loose_stone ls
       ON i.lot_id = ls.lot_id
       LEFT JOIN white_diamond wd
       ON ls.lot_id = wd.lot_id
       LEFT JOIN colored_diamond cd
       ON ls.lot_id = cd.lot_id
       LEFT JOIN colored_gem_stone cgs
       ON ls.lot_id = cgs.lot_id
       LEFT JOIN jewelry j
       ON i.lot_id = j.lot_id
 GROUP BY item_tp
 ORDER BY total_count DESC;
```

SQL (3)

'Keep track of availability and location' trigger (part 1)

```
CREATE OR REPLACE FUNCTION trig_a_i_keep_track_responsible_office()
    RETURNS TRIGGER AS
$$
DECLARE
    counterpart_col_name text;
    item_availability bool;
    dynamic_sql text;
BEGIN
    IF TG_TABLE_NAME IN ('return_memo_out', 'back_from_factory', 'back_from_lab') THEN
        counterpart_col_name := 'to_counterpart_id';
        item_availability := TRUE;
    ELSIF TG_TABLE_NAME IN ('return_memo_in', 'memo_out', 'transfer_to_factory', 'transfer_to_lab', 'sale') THEN
        counterpart_col_name := 'from_counterpart_id';
        item_availability := FALSE;
    ELSE -- transfer_to_office
        counterpart_col_name := 'to_counterpart_id';
        item_availability := TRUE;
    END IF;
    -- part 2 next slide
    RETURN new;
END;
$$ LANGUAGE plpgsql;
```

SQL (4)

'Keep track of availability and location' trigger (part 2)

```
CREATE OR REPLACE FUNCTION trig_a_i_keep_track_responsible_office()
    RETURNS TRIGGER AS
$$
DECLARE
    counterpart_col_name text;
    item_availability bool;
    dynamic_sql text;
BEGIN
    -- part 1 prev slide
    dynamic_sql := FORMAT(
        $sql$
        UPDATE diamonds_are_forever.item
            SET responsible_office_id = (
                SELECT %I FROM diamonds_are_forever.action WHERE action_id = $1
            ), updated_at   = NOW(), is_available = $2
        WHERE lot_id IN (
            SELECT lot_id
            FROM diamonds_are_forever.action_item WHERE action_id = $1
        )
        $sql$, counterpart_col_name); -- part 3 on the next slide
END; $$ LANGUAGE plpgsql;
```

SQL (5)

`Keep track of availability and location` trigger (part 3)

```
CREATE OR REPLACE FUNCTION trig_a_i_keep_track_responsible_office()
    RETURNS TRIGGER AS
$$
DECLARE
    counterpart_col_name text;
    item_availability bool;
    dynamic_sql text;
BEGIN
    -- part 2 on the prev slide
    EXECUTE dynamic_sql
        USING NEW.action_id, item_availability;

    RETURN new;
END; $$ LANGUAGE plpgsql;
```

SQL (6)

`Keep track of availability and location` trigger (part 4)

```sql
CREATE TRIGGER responsible_office_after_purchase_trigger
    AFTER INSERT ON purchase FOR EACH ROW
EXECUTE FUNCTION trig_a_i_keep_track_responsible_office();

CREATE TRIGGER responsible_office_after_transfer_to_factory_trigger
    AFTER INSERT ON transfer_to_factory FOR EACH ROW
EXECUTE FUNCTION trig_a_i_keep_track_responsible_office();

CREATE TRIGGER responsible_office_after_back_from_factory_trigger
    AFTER INSERT ON back_from_factory FOR EACH ROW
EXECUTE FUNCTION trig_a_i_keep_track_responsible_office();

CREATE TRIGGER responsible_office_after_sale_trigger
    AFTER INSERT ON sale FOR EACH ROW
EXECUTE FUNCTION trig_a_i_keep_track_responsible_office();
```

SQL (7)

Make a purchase procedure (part 1)

```
CREATE OR REPLACE PROCEDURE pcd_create_purchase(
    IN p_supplier_id           INT,
    IN p_office_id             INT,
    IN p_purchase_num          TEXT,
    IN p_stock_name            TEXT,
    IN p_origin                TEXT,
    IN p_item_type             diamonds_are_forever.item_category,
    IN p_price                 NUMERIC,
    IN p_currency_code         diamonds_are_forever.code,
    OUT v_lot_id INT
)
LANGUAGE plpgsql
AS $$
DECLARE
    v_action_id INT;
BEGIN
    INSERT INTO diamonds_are_forever.item(
        stock_name,
        supplier_id, origin,
        responsible_office_id, item_type
    ) VALUES (
        p_stock_name, p_supplier_id,
        p_origin, p_office_id, p_item_type
    ) RETURNING lot_id INTO v_lot_id;
    -- part 2 next slide
END;
$$;
```

SQL (8)

Make a purchase procedure (part 2)

```
CREATE OR REPLACE PROCEDURE pcd_create_purchase(
    IN p_supplier_id           INT,
    IN p_office_id             INT,
    IN p_purchase_num          TEXT,
    IN p_stock_name            TEXT,
    IN p_origin                TEXT,
    IN p_item_type             diamonds_are_forever.item_category,
    IN p_price                 NUMERIC,
    IN p_currency_code         diamonds_are_forever.code,
    OUT v_lot_id INT
)
LANGUAGE plpgsql
AS $$
DECLARE
    v_action_id INT;
BEGIN
    -- part 1 prev slide
    INSERT INTO diamonds_are_forever.action(
        from_counterpart_id, to_counterpart_id, action_category, remarks
    ) VALUES (
        p_supplier_id, p_office_id, 'purchase', 'Purchase created via procedure'
    ) RETURNING action_id INTO v_action_id;
    --part 3 next slide
END;
$$;
```

SQL (9)

Make a purchase procedure (part 3)

```
CREATE OR REPLACE PROCEDURE pcd_create_purchase(
    IN p_supplier_id           INT,
    IN p_office_id             INT,
    IN p_purchase_num          TEXT,
    IN p_stock_name            TEXT,
    IN p_origin                TEXT,
    IN p_item_type             diamonds_are_forever.item_category,
    IN p_price                 NUMERIC,
    IN p_currency_code         diamonds_are_forever.code,
    OUT v_lot_id INT
)
LANGUAGE plpgsql
AS $$
DECLARE
    v_action_id INT;
BEGIN
    -- part 2 prev slide
    INSERT INTO diamonds_are_forever.purchase(action_id, purchase_num)
    VALUES (v_action_id, p_purchase_num);
    INSERT INTO diamonds_are_forever.action_item(
        action_id, lot_id, price, currency_code)
    VALUES (v_action_id, v_lot_id, p_price, p_currency_code);
END;
$$;
```

Tech stack

- Frontend: Streamlit
- Database: PostgreSQL 18
- Python: 3.13
- Package Manager: uv
- Container: Docker

Demo

Challanges

- Design document to Conceptual schema
- UI design (what a person who is going to be use it will need)
- Correcting item's data when item has undergone at least one action
- Correcting action's data if the it isn't the most recent action

Planed features v.s. Implemented features (1)

Inventory operations:	
Purchase: registering new items	yes
Memo Out: sending items to clients for approval	yes
Return Memo Out: returning items sent for approval	yes
Sale: registering sales with payment details	yes
Transfer to Lab: sending to laboratory for certification	yes
Back from Lab: returning items with new certificates	yes
Transfer to Factory: sending to factory for processing	yes
Back from Factory: returning items with new dimensions	yes
Memo In: receiving goods on consignment	yes
Return Memo In: returning consigned goods	yes
Transfer To Office: transfering goods between offices	yes

Planed features v.s. Implemented features (2)

Consultation and Reporting	yes
Complete view by item type	yes
Automation location status calculation	yes
Complete action history per lot	yes
Inventory by type with statistics	yes
Filtering by status, availability, location	yes

Planed features v.s. Implemented features (2)

Traceability	yes
Timestamped recording of all actions	yes
Linking between actions (returns referencing original transfers)	yes
Complete item history via `get_item_history` function	yes
Update logs in `action_update_log`	yes

Planed features v.s. Implemented features (2)

Financial reports: revenue analysis by period	no
Grouping by clients: revenue by client	no
Grouping by category: revenue by item type	no
Revenue charts: temporal visualizations	no

Planed features v.s. Implemented features (2)

Application	
Registering new white/colored diamond via purchase	yes
Registering new sale	yes
Transfering item to lab/factory/office	yes
Returning item from lab/factory/office	yes
Inspecting item (white/colore diamond, gemstone, jewelry)	yes
Inspecting certificate details	no
Basic authentication (via email)	yes
Counterpart management interface : add/modify suppliers, clients, labs	no
Employee account management: add/modify user accounts	no
PDF generation of administrative reports	no
Verification that user has filled all the requirement fields	no
Registering Memo In, Memo Out and corresponding returns	no

Conclusion / Result

1. **Full-stack database application development**
2. **Database design**
3. **Production-ready deployment**

Thank you for your attention !